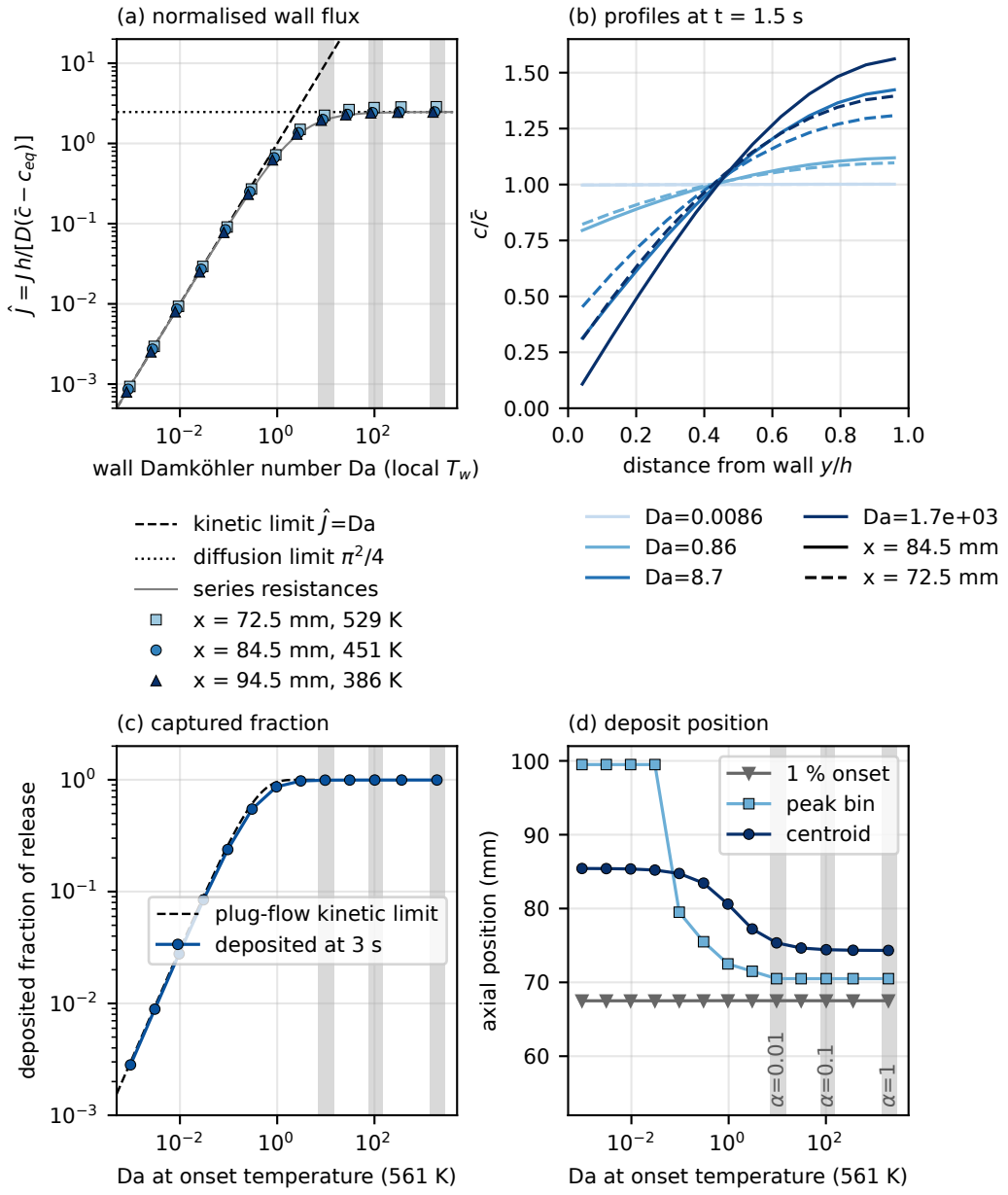
(a) normalised wall flux
$\hat{J} = Jh/[D(\tilde{c} - c_{eq})]$
wall Damköhler number Da (local T_w)
kinetic limit $\hat{J}$=Da
diffusion limit $\pi^2/4$
series resistances
x = 72.5 mm, 529 K
x = 84.5 mm, 451 K
x = 94.5 mm, 386 K
(b) profiles at t = 1.5 s
$c/\tilde{c}$
distance from wall y/h
Da=0.0086
Da=0.86
Da=8.7
Da=1.7e+03
x = 84.5 mm
x = 72.5 mm
(c) captured fraction
deposited fraction of release
plug-flow kinetic limit
deposited at 3 s
Da at onset temperature (561 K)
(d) deposit position
axial position (mm)
1 % onset
peak bin
centroid
α=0.01
α=0.1
α=1
Da at onset temperature (561 K)